

Analyzing Company Stock Behavior Based on News Article Headline Sentiment Using FinBERT

Dylan Guzman

DATA 580F AI Driven Decision Support

Dr. Hafiz Munsub Ali

May 12, 2026

Abstract

This study aims to determine how well sentiment analysis scores obtained from finBERT can predict company stock price change compared to baseline forecasting models. The sentiment scores are obtained from every news article headline on the first page of google when either the name of a fortune 500 company is searched for, or when the word “Trump” is searched for. Six types of GRU models were then made for each company in the study. A baseline using only previous price change as inputs, one model using previous company sentiment scores as inputs, one model using previous Trump sentiment scores as inputs, two models using each type of sentiment and previous price change as inputs, and one model using all three variables as inputs. By comparing RMSE, MSE, and overall portfolio growth when investing using the prediction from each model the study concludes that finBERT sentiment scores from company headlines do not aid in predicting stock price change, but sentiment scores from Trump articles can aid in predicting stock price change.

1. Introduction

The stock market has proven to be affected by the sentiment of financial news sources due to the influence news sources have over investors [1]. Due to the vast amount of articles and blogs that get posted in the current day and age, it has been difficult to accurately quantify the effect that financial news has on company stock prices. The recent technological advancements to large language models and transformers has made it easier to collect and analyze this data, but has proven to be ineffective in accurately predicting whether articles have a positive or negative sentiment toward a company [2]. Additionally, LLMs cannot adjust to the volatile nature of the stock market because of the lack of market patterns [3]. To get around this problem, some studies have manually annotated articles to ensure the models being used are not misrepresenting articles [4]. However, this can be a time consuming and expensive tradeoff.

Despite the limitations of LLMs, one has come to the forefront when it comes to examining the sentiment from financial texts. FinBERT is a variation of Google’s BERT which has been trained on thousands of financial texts in order to capture the sentiment from these texts. While finBERT has achieved accuracies of nearly 90% [5] and proven to be the premier LLM for sentiment analysis of financial texts, it is unclear if finBERT’s can be used to predict company stock price. This study aims to determine if news article headlines scored by finBERT can be used as an effective predictor for company stock price.

2. Literature Review

One study uses LLMs such as finBERT and Llama 3.1 to gather the sentiment from news articles about the top 30 companies from the NASDAQ to capture the effect articles have on company stock price [6]. The study then employed three different investment strategies, and found that using the sentiment found, only a long-term strategy was effective. Another study wants to make advancements in sentiment detection with the use of LLMs, in this case finBERT [7]. The researchers collect sentiment from news articles using finBERT and predict stock prices which are compared to simulated market results to look for areas of improvement. A third study uses the LLMs finBERT and VADER to collect sentiment analysis from news articles about companies. The sentiment analysis is then used by neural network modeling to predict the next day movement of company stock prices [8]. Lastly, a similar study also aims to predict the next day movement of company stock prices using sentiment analysis. However, this study uses a wider range of LLMs, including finBERT, DeBERTa, and RoBERTa [9].

Similarly to all of these studies, I also plan to use finBERT to gather sentiment analysis from news articles. However, I plan to focus solely on the article headlines, while these studies use the entire article. Additionally, the first study only uses 30 companies where 55 will be used in this study. The second study only uses articles from “reputable” sources where this study will have no limitations on articles used. Finally, the last two studies use directional accuracy as its only evaluation metric. This study plans on using directional accuracy, as well as other evaluation metrics such as RMSE and MASE.

The stock market as a whole moves according to financial sentiment whether that be from news articles or social media, and it is important to quantify the sentiment to predict stock market behavior. One study aims to do this using the content from financial news articles and Twitter posts in conjunction with word2vec and BERT sentiment indexes for sentiment analysis and convolutional neural networks for forecasting [10]. Other areas of sentiment analysis include social media posts, as they also have an effect on stock market prices and should be quantified so investors know the extent of this effect. A study utilizes parts-of-speech tagging for sentiment analysis of twitter posts along with linear regression and deep neural networks for stock price prediction [11]. A third study that analyzes social media posts and financial news utilizes advanced forecasting models such as XGBoost and BiLSTM models to forecast stock price, along with finBERT to collect sentiment analysis from media [12].

While these studies attempt to forecast stock market behavior through sentiment analysis, they differ from this study because they are analyzing social media posts where this study analyses news article headlines. Also, the first and third study examines the effect of sentiment on the stock market as a whole. This study plans on doing that as well, but it also will look at how sentiment affects stock price movement on a company-level. Lastly, The second study mentioned chooses to only examine 5 companies very closely where this study has 55 in its dataset.

Sentiment analysis may have a greater impact on companies of a specific sector, and is looked into by a study that researches sentiment analysis among energy companies [13]. It uses finBERT for sentiment analysis and long-short-term memory models for prediction among seven energy companies in the New York Stock Exchange over a ten year period. This study uses a centralized approach, looking at a few specific companies over a long period of time, while my objective is to look at a broad range of companies over a short period of time. It is important to explore sentiment analysis from social media posts using differing collection methods. For instance, one study used graph neural networks with natural language processing methods to collect sentiment analysis and long-short-term memory models for prediction [14]. My objective differs as I am exploring sentiment analysis from news articles rather than social media posts.

3. Methodology

Data Collection & Cleaning

This study aims to quantify the sentiment of news article headlines on company stock price. By using BeautifulSoup for web scraping and ThreadPoolExecutor to scrape multiple titles simultaneously, every article title on the first page of Google's news section was collected for every day of 2025 when each Fortune 500 company was searched. For instance, on June 16, 2025 an article titled “Tesla leads U.S. sales of EVs lower in April, marking the first annual drop in over a year” is a result when “Tesla” is searched in Google’s search engine, thus is being used as data for this study. Additionally, every article on the first page of Google’s news section was collected for every day of 2025 when the word “Trump” was searched, in order to determine if news surrounding the president affects company stock price. To gather the company stock price, this study will be using Yahoo Finance. Specifically, the closing price will be gathered from the yf python library for every day in 2025 for each company being analyzed.

Fortune 500 companies were chosen due to their high reputation, making them more likely to have plenty of news articles written about them. That being said, most of these companies have days where there are no articles written about. For example, when “HF Sinclair” is searched, there are no results from September 14th to 17th, meaning no articles were published about this company during that time frame. To ensure that only companies with a sufficient amount of articles written about them are being used, only companies with at least 330 days of 5 or more articles written about them were used. This left only 55 publicly traded companies for the study.

To quantify the sentiment of news article titles, this study will be using finBERT, an LLM pre-trained on financial texts that assigns text probabilities based on how likely it is to be positive, negative, or neutral [15]. The main metric used in this study is acquired by applying this LLM to every news article title, then taking the difference of the average positive and average negative scores for each day among each company. This metric will be referred to as “sentiment score” in the study. For days where a company had less than 5 articles written about them, the most recent previous sentiment score was used.

Stock Performance Analysis & Evaluation

To determine if the sentiment gathered from news article headlines using finBERT is effective at predicting company stock price, a baseline first has to be established. For this study the baseline will be a gated recurrent unit (GRU) model using a rolling seven day window as the input. GRUs are a type of neural network architecture that forgets information that is no longer useful to avoid

an overloading of information while retaining useful information to be used for prediction [16]. GRUs are great for sequential data and will be a good fit for evaluating stock prices over time. Root mean squared error (RMSE), mean absolute scaled error (MASE), and directional accuracy will be used to evaluate model performance. RMSE measures the distance between predicted values and actual values, while penalizing larger differences and using the same units as the target variable. Time series errors typically follow a normal distribution, which is the distribution best suited for RMSE [17]. MASE measures the ratio of the mean absolute errors for a forecasting model and the mean absolute error of the one-lag naive forecast and is referred to as the best way to compare forecasting models [18]. Directional accuracy in this context is the percentage of predictions that each model correctly predicts whether company stock price will increase or decrease.

Each company will have six models made for them in this study, each taking different sets of inputs over the previous seven days to predict the company's stock price change of the next day. The first is the baseline, which will use the daily percent change of stock price as its input. The second model will use the sentiment scores from company articles from the previous seven days as its input. The third model will use both the percent change of stock price and the sentiment scores from company articles as inputs. The fourth model will use the sentiment scores from Trump articles as an input. The fifth model will use both the sentiment scores from Trump articles and price change as inputs. The sixth and final model will use all three variables previously mentioned as inputs. The objective is to determine if models two through six (which will be commonly referred to as the "test models") yield better results than the baseline model. To compare which models work best, the study will compare the previously mentioned evaluation metrics across different model types, and will use paired one tail T-tests to determine if the average difference in metrics are statistically significant.

These models will also be used to create a portfolio using an investment strategy based on each model. If a model predicts a price increase greater than 0.1%, the portfolio will long the stock. Conversely, if a model predicts a price decrease greater than 0.1%, the portfolio will short the stock. Any values in between, and the portfolio will neither long nor short the stock. To evaluate the success of the portfolios, this study will be using spearman rank correlation and overall portfolio growth. Spearman rank correlation measures the direction of an increasing or decreasing relationship between two or more variables, and will be useful to tell if a portfolio is always increasing thus successful. For all models and portfolios, the first nine months of 2025 will be used as training data, and the last three months will be used as testing data.

4. Results

It's hard to assess which type of model is best across 55 different companies, but a good starting point is to look at the average evaluation metrics for each model type. In table 1, these averages are displayed for each model type, with the best value in each column being bolded. (Note that “DA” is short for directional accuracy, “SR” is short for spearman rank, and “PR” is short for portfolio return).

Table 1:

<i>model</i>	Average RMSE	Average MASE	Average DA	Average SR	Average PR
combined	0.1128	5.0859	0.4859	0.0141	-0.0121
price_and_sentiment	0.0534	2.3017	0.4892	-0.0039	-0.0099
price_only	0.0375	1.5222	0.4841	-0.0414	-0.0013
sentiment_only	0.0414	1.6965	0.4763	0.0220	0.0011
trump_and_price	0.0477	2.0694	0.4846	0.0263	0.0133
trump_only	0.0330	1.3313	0.5029	0.0827	0.0593

Across the board, the “trump only” model is best in every average evaluation metric, and is the only model type that outperforms the baseline in both RMSE and MASE. By using T-tests, it can be determined whether these differences between the baseline model and the test models are significant or not. Below, table 2 displays the P-value of one sided paired t-tests using the baseline model and each model shown in the model column with the significant value at the 0.05 level being in bold. The tests assume the null hypothesis that the baseline model will have the same average scores for each metric compared to the other models. The alternate hypothesis is that the baseline model will score better on average than the other models for each metric.

Table 2:

Model	RMSE	MASE	DA	SR	PR
Trump_only	0.0000	0.0000	0.0594	0.0000	0.0184
Trump_Price	0.0000	0.0000	0.4798	0.0002	0.3038
Sentiment_only	0.0031	0.0011	0.2692	0.0163	0.4685
Sentiment_Price	0.0000	0.0000	0.3137	0.0580	0.3780
Combined	0.0000	0.0000	0.4378	0.0069	0.3373

Table 2 further proves the claim that the “trump only” models outperformed the baseline price models. All average evaluation metrics for the “trump only” models are statistically significantly better than the average evaluation metrics for the baseline models, aside from directional accuracy. As for the other test models, table 2 shows they had statistically significantly worse average RMSE, and MASE values than the baseline model.

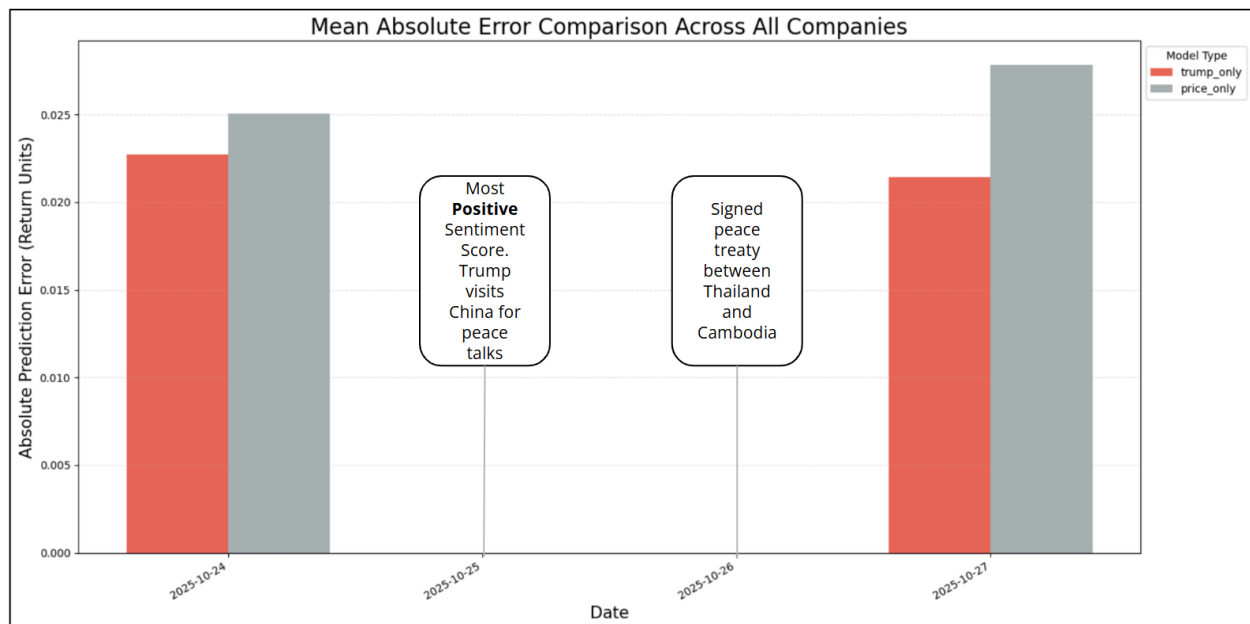
To further demonstrate how the models performed comparatively, the percentage of companies that outperformed the baseline in each metric was calculated and shown in table 3. As expected, the majority of “trump only” models performed better than the baseline model across every metric, and had the highest value among each model type for every metric. Most notably, the “trump only” models had the highest percentage of companies with a higher directional accuracy, which is crucial for financial forecasting.

Table 3:

Model	RMSE	MASE	DA	SR	PR
trump_only	80.0%	80.0%	60.0%	76.4%	67.3%
trump_and_price	9.1%	0.0%	41.8%	67.3%	50.9%
sentiment_only	32.7%	27.3%	38.2%	67.3%	47.3%
sentiment_and_price	1.8%	0.0%	54.5%	61.8%	50.9%
combined	0.0%	0.0%	49.1%	63.6%	54.5%

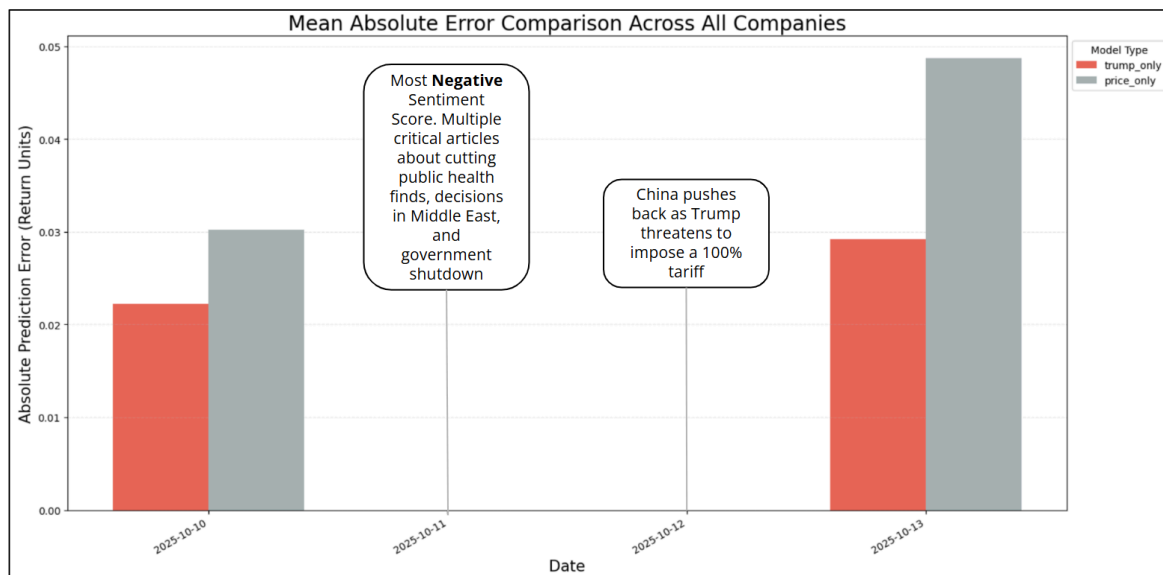
Graphing the average absolute error across all companies for extreme sentiment values can help provide an understanding as to why the “trump only” models have outperformed the baseline models. Graph 1 displays the average absolute error for the “trump only” model and the baseline model after the day with the most positive sentiment score in the test data, October 25, 2025. On this day, which was a Saturday, Donald Trump left to visit Asia to have peace talks with Thailand and Cambodia, and the following day signed a peace treaty with both countries. When the market reopened on Monday, the “trump only” models were able to utilize the positive sentiment to achieve an error of 0.5% less than the baseline error.

Graph 1:



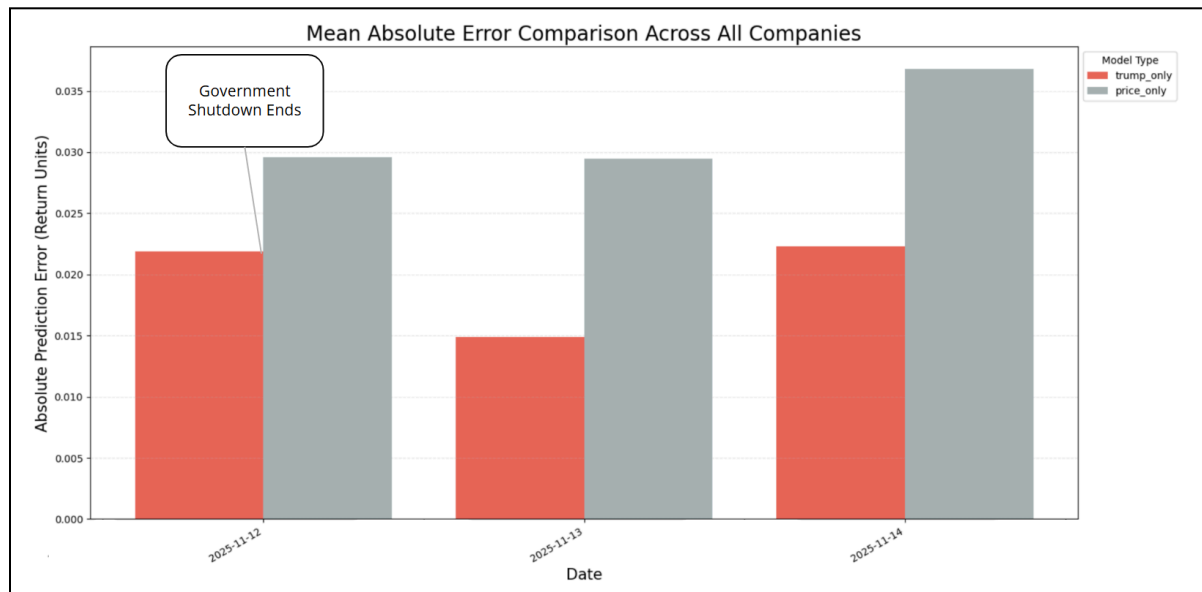
On the other side of the sentiment spectrum, the “trump only” models outperform the baseline models by roughly 2% on average after the day with the most negative sentiment score and can be seen in graph 2. On October 11, 2025 which was also a Saturday, multiple articles were written criticizing the president for cutting public health programs, his decisions toward conflicts in the Middle East, and for the government being shut down. On Sunday, China stated that they would retaliate if Trump imposed the 100% tariff that he previously threatened. On Monday, the “trump only” model had an average absolute error of just under 3%, where the baseline model had an average absolute error of almost 5%.

Graph 2:



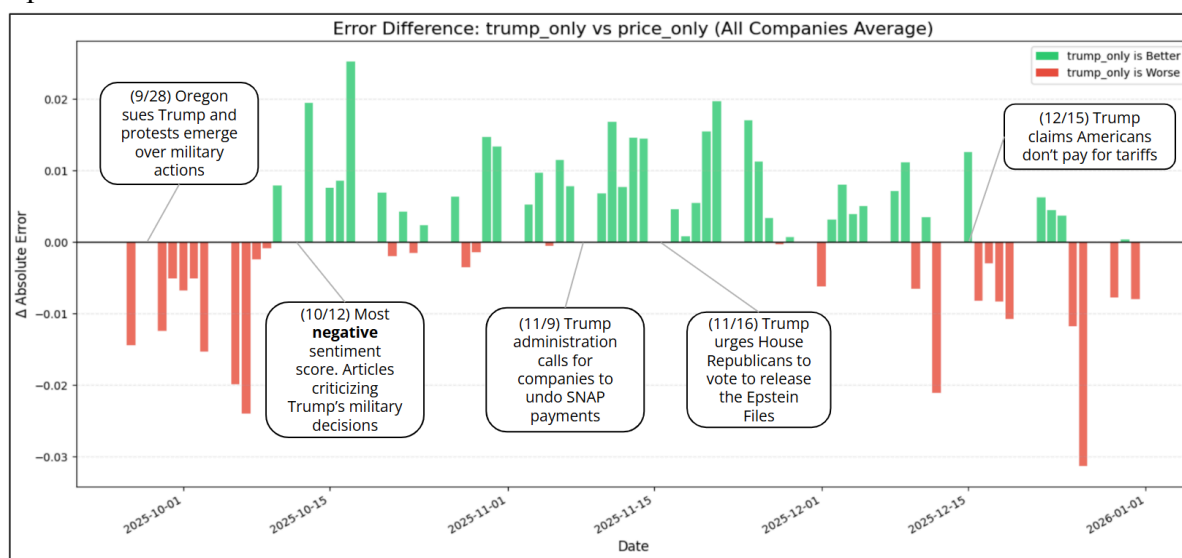
Additionally, the “trump only” model seemed to outperform the baseline model during major political events. For instance, graph 3 displays the mean absolute error for the two days after the 2025 government shutdown ended, on November 12. The sentiment from the Trump articles can be used to more accurately make predictions, resulting in the “trump only” models having less error than the price models.

Graph 3:



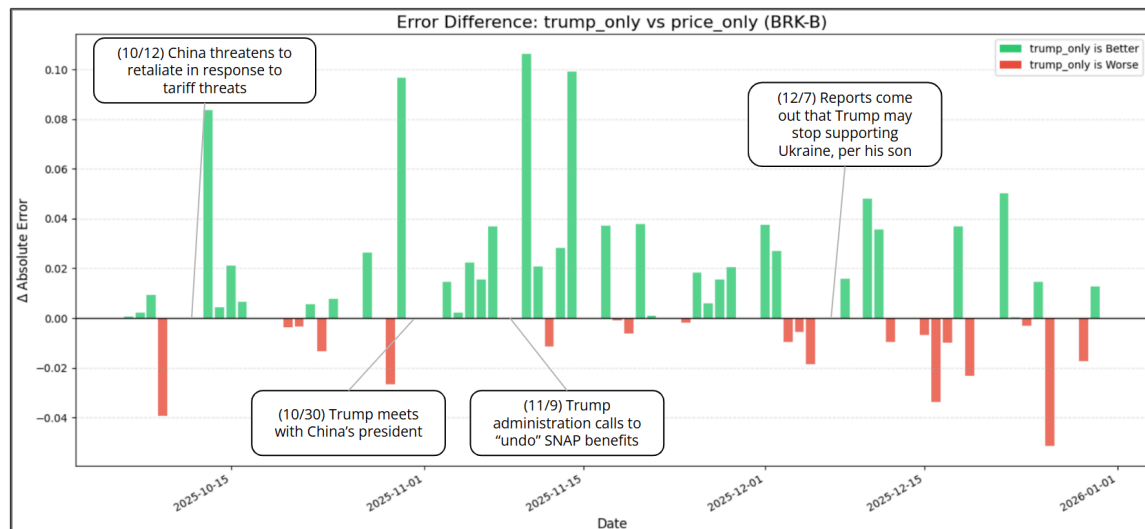
With the context of how the models perform during the extremes, it's easier to understand how the models perform over a longer period of time. Graph 4 presents the average difference in error across all companies between the “trump only” models and the baseline models where positive green bars mean the “trump only” model had less prediction error and the negative red bars mean the baseline model had less prediction error. The graph also highlights significant events that preceded several consecutive days of positive or negative model performance for the “trump only” model. Graph 4 clearly depicts how the “trump only” models perform better than the baseline models, especially after instances of criticism or articles relating to Trump’s image such as articles pertaining to the Epstein Files.

Graph 4:



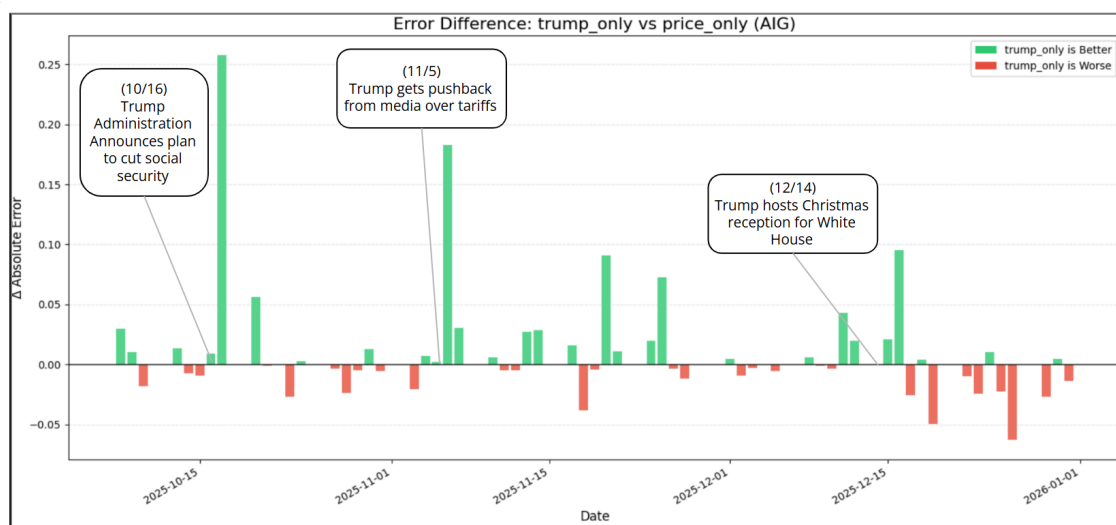
The same concepts apply to the company-level. Berkshire-Hathway had the greatest difference in MASE between the “trump only” model and baseline model and is displayed in graph 5, along with major events followed by success from the “trump only” model. Since Berkshire-Hathway is a conglomerate of over 50 companies, it makes sense as to why it is heavily affected by sentiment from Trump articles.

Graph 5:



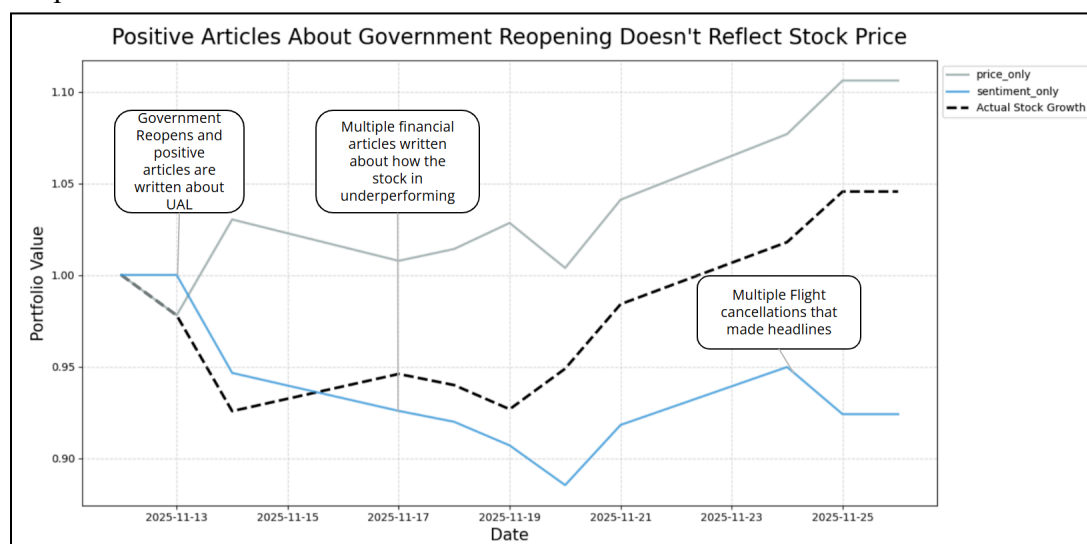
Another company that found greater success with the “trump only” model was American International Group (AIG). Similar to the last two graphs, graph 6 depicts the difference in error between models for AIG. There are a couple instances of the “trump only” model greatly outperforming the baseline model, which are highlighted by the news that preceded them. Aside from that, the two models performed similarly.

Graph 6:



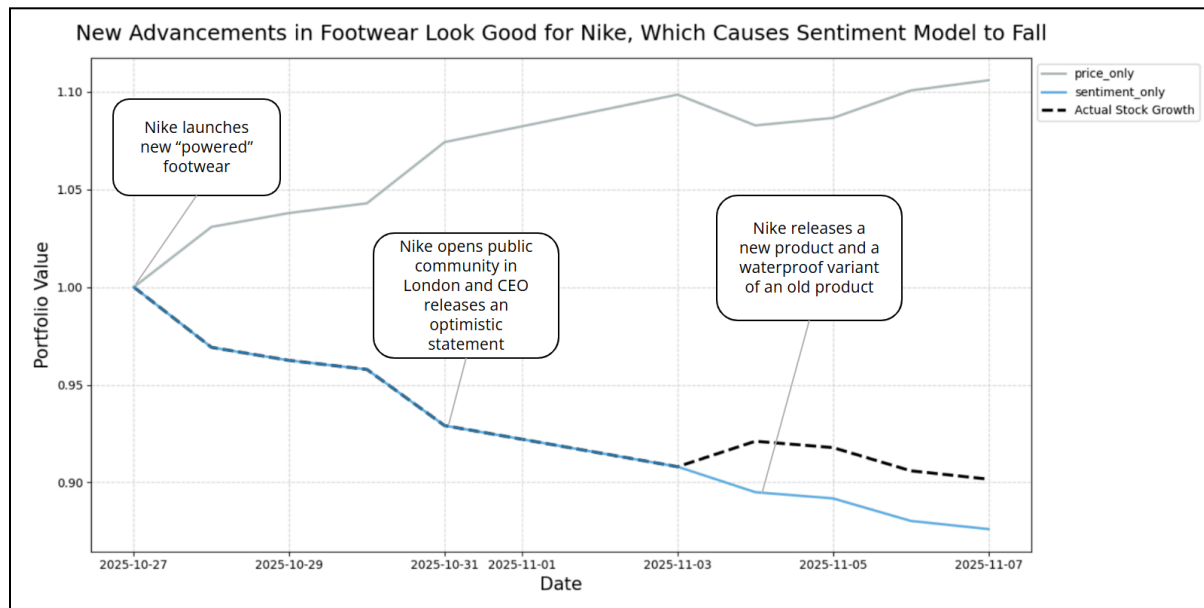
As for the models using sentiment relating to the company, graphing some results may lead to insights as to why the baseline price models outperform these sentiment models. For instance in graph 7, which shows the portfolio returns based on investment strategies using different model types for United Airlines Holding Inc. (UAL), the baseline model greatly exceeds the value of the “sentiment only” model. On November 13, 2025 several articles were written about the government reopening, and how it would positively affect airlines including UAL, causing the “sentiment_only” model to buy the stock. However, the actual stock price did not increase over the next few days, leading the “sentiment only” model to suffer. Additionally, on November 24, 2026 multiple articles about flight cancellations were released, causing the “sentiment only” model to have the opposite effect and short the stock, when in reality the stock price increased, causing the model’s value to fall even more.

Graph 7:



Graph 8 showcases another example of the baseline model surpassing the value of the “sentiment only” model. From October 27, 2025 to November 7, 2025, Nike has a series of product releases and innovations that lead to high sentiment scores and the “sentiment only” model to buy Nike stock. However, Nike’s stock would fall nearly 10% during this time period, while the “price only” model can detect the stock is falling and successfully short the stock to increase its value by over 10%.

Graph 8:



5. Conclusion

The goal of this study was to determine if the sentiment analysis of news article headlines gathered using the finBERT LLM could be used as a predictor for company stock price. Ultimately, the findings suggest that the finBERT sentiment analysis from company specific headlines actually hurts model performance, and GRUs performed better using just previous stock movement as a predictor. However, when sentiment analysis was gathered from articles about Donald Trump and used as a predictor it was more successful than baseline GRU models using only previous stock price movement as a predictor.

This should not be a conclusive result that finBERT's sentiment analysis from company articles is definitively ineffective in predicting company stock price and that sentiment analysis from Trump articles are effective at the same task. This study only looked at article titles and a seven day window of previous sentiment score and stock price movement. Further research should examine the effects of the content of articles as well as different time periods of previous stock price change and sentiment.

6. References

1. Ehsan, Aqsa, et al. "Financial News Sentiment Analysis Using NLP and Machine Learning for Asset Price Prediction: A Systematic Review." *VFAST Transactions on Software Engineering*, vol. 13, no. 3, 30 Sept. 2025, pp. 279–308, vfast.org/journals/index.php/VTSE/article/view/2165, <https://doi.org/10.21015/vtse.v13i3.2165>.
2. Kang, Ji-Won, and Sun-Yong Choi. "Comparative Investigation of GPT and FinBERT's Sentiment Analysis Performance in News across Different Sectors." *Electronics*, vol. 14, no. 6, 10 Mar. 2025, pp. 1090–1090, <https://doi.org/10.3390/electronics14061090>. Accessed 9 Apr. 2025.
3. Mu, Zongshen, et al. "Exploring Financial Sentiment Analysis via Fine-Tuning Large Language Model and Attributed Graph Neural Network." *Neural Networks*, vol. 199, 22 Jan. 2026, p. 108620, www.sciencedirect.com/science/article/abs/pii/S0893608026000821, <https://doi.org/10.1016/j.neunet.2026.108620>.
4. Muhammad, Iftikhar, and Marco Rospocher. "On Assessing the Performance of LLMs for Target-Level Sentiment Analysis in Financial News Headlines." *Algorithms*, vol. 18, no. 1, 13 Jan. 2025, p. 46, <https://doi.org/10.3390/a18010046>. Accessed 16 Jan. 2025.

5. Baghavathi Priya, S, et al. "Advanced Financial Sentiment Analysis Using FinBERT to Explore Sentiment Dynamics." *2025 3rd International Conference on Intelligent Data Communication Technologies and Internet of Things (IDCIoT)*, 5 Feb. 2025, pp. 889–897, ieeexplore.ieee.org/abstract/document/10915080, <https://doi.org/10.1109/idciot64235.2025.10915080>. Accessed 1 June 2025.
6. Mun, Yejoon, and Namhyoung Kim. "Leveraging Large Language Models for Sentiment Analysis and Investment Strategy Development in Financial Markets." *Journal of Theoretical and Applied Electronic Commerce Research*, vol. 20, no. 2, 20 Apr. 2025, p. 77, <https://doi.org/10.3390/jtaer20020077>. Accessed 9 May 2025.
7. Qi, Ruoyu. "Financial News Sentiment Analysis and Market Sentiment Prediction Based on Large Language Models." *Journal of Computer, Signal, and System Research*, vol. 2, no. 6, 2025, pp. 22–31, www.gbspress.com/index.php/JCSSR/article/view/473, <https://doi.org/10.71222/y0vdp428>.
8. Dipura, Ryan Wira Ariza, et al. "Stock Market Trend Prediction for next Day Direction Using Sentiment Analysis with FinBERT." *2025 11th International Conference on Communication and Signal Processing (ICCSP)*, 5 June 2025, pp. 1631–1636, ieeexplore.ieee.org/abstract/document/11088503?casa_token=ImaW33eLBukAAAAA:QrKt5Uhrsx8FubG9-YxQ3FkH0Lqq6qfQbCKaaktUGOgjd-rfCIESZbsv05TBbeVoP2lgOkI, <https://doi.org/10.1109/iccsp64183.2025.11088503>. Accessed 30 Apr. 2026.
9. Siala, Walid, et al. "Impact of LLMs News Sentiment Analysis on Stock Price Movement Prediction." *ArXiv.org*, 2026, arxiv.org/abs/2602.00086.
10. Yang, Wei, et al. *Research on Stock Market Sentiment Analysis and Prediction Method Based on Convolutional Neural Network*. 22 Apr. 2025, pp. 91–96, <https://doi.org/10.1145/3747227.3747241>.
11. Saravanos, Christina, and Andreas Kanavos. "Forecasting Stock Market Volatility Using Social Media Sentiment Analysis." *Neural Computing and Applications*, 13 Dec. 2024, <https://doi.org/10.1007/s00521-024-10807-w>.
12. H, Niroshini Infantia, et al. "A Hybrid BiLSTM–Transformer–FinBERT–XGBoost Framework for End-of-Day Stock Prediction and Trading Recommendations." *2026 International Conference on Innovative Computing, Intelligent Communication and Smart Electrical Systems (ICSES)*, 29 Jan. 2026, pp. 1–10, ieeexplore.ieee.org/abstract/document/11479182?casa_token=onO8-W8RcqqAAAAA:IQ5F7U6dAnyTUWdVpGn5xT2h7BrOy4lzdJ1E_fDdwI0UEt78icZ4_ysEnU5aJlyFD0ZhayEE, <https://doi.org/10.1109/icses66558.2026.11479182>. Accessed 30 Apr. 2026.
13. Lee, Chi-Yuan, and Eva Anderl. "Does Business News Sentiment Matter in the Energy Stock Market? Adopting Sentiment Analysis for Short-Term Stock Market Prediction in the Energy Industry." *Frontiers in Artificial Intelligence*, vol. 8, 23 July 2025, pmc.ncbi.nlm.nih.gov/articles/PMC12327087/, <https://doi.org/10.3389/frai.2025.1559900>.

14. Zhang, Pengju, et al. "GNN-Based Social Media Sentiment Analysis for Stock Market Forecasting and Trading." *Expert Systems with Applications*, vol. 291, Oct. 2025, p. 128425, <https://doi.org/10.1016/j.eswa.2025.128425>. Accessed 1 July 2025.
15. Huang, Allen H., et al. "FinBERT: A Large Language Model for Extracting Information from Financial Text†." *Contemporary Accounting Research*, vol. 40, no. 2, 29 Sept. 2022, <https://doi.org/10.1111/1911-3846.12832>.
16. Chung, Junyoung, et al. "Empirical Evaluation of Gated Recurrent Neural Networks on Sequence Modeling." *ArXiv.org*, 2014, arxiv.org/abs/1412.3555.
17. Hodson, Timothy O. "Root-Mean-Square Error (RMSE) or Mean Absolute Error (MAE): When to Use Them or Not." *Geoscientific Model Development*, vol. 15, no. 14, 19 July 2022, pp. 5481–5487.
18. Franses, Philip Hans. "A Note on the Mean Absolute Scaled Error." *International Journal of Forecasting*, vol. 32, no. 1, Jan. 2016, pp. 20–22, <https://doi.org/10.1016/j.ijforecast.2015.03.008>. Accessed 19 Dec. 2019.